

Solution Method

This section describes the method used to solve the equations. The method is based on the simultaneous solution of all equations using Newton's method. In compositional simulation, a volume constraint equation is normally also part of the equation set:

$$\psi_p \equiv \sum_q \left(\frac{N_q^{n+1}}{\rho_q^{n+1}} \right) - \phi^{n+1} = 0, \quad \mathbf{q} = \mathbf{g}, \mathbf{w} \quad (\text{D12.1})$$

Equation (D12.2) equates the sum of phase volumes per unit reservoir volume to the pore volumes. Let $\vec{\psi}$ be the vector of the material balance equations for all gaseous and aqueous components (Equations (D10.1) and (D10.2)). Let $\vec{\psi} = \mathbf{E}\vec{\psi}$ be the n_{cp} reduced set of "primary component" material balance equations ($n_{cp} = n_{aq} - R_{aq}$ with $n_{aq} = n_c + n_a$).

The following equations are taken as primary equations:

The volume constraint equation (Equation (D12.1)),

The $> n_{cp}$ reduced aqueous component material equations $\vec{\psi}$,

The n_c phase equilibrium equations (Equation (D2.1)),

The R_{aq} chemical equilibrium equations (Equation (D6.1)),

The n_m mineral balance equations (Equation (D10.3)).

In total, there are $(1 + n_{cp} + n_c + R_{aq} + n_m)$ equations. The primary unknowns are:

Pressure,

The summed mole numbers of gaseous primary components in the gas and aqueous phases,

$$N_i, i = 1, \dots, n_c,$$

The mole numbers of aqueous components (excluding the gaseous components that are soluble in the aqueous phase),

$$N_j, j = n_c + 1, \dots, n_c + n_a,$$

The mole numbers of minerals,

$$N_k, k = n_c + n_a + 1, \dots, n_{ct},$$

The aqueous mole numbers of gaseous components soluble in the aqueous phase,

$$N_{i,w}, i = 1, \dots, n_c.$$

Let $\vec{\psi}$ be the vector comprising all equations for all grid blocks and $\vec{\xi}$ the vector of all primary variables. The well constraint equations and flowing bottomhole pressures are also included in $\vec{\psi}$ and $\vec{\xi}$, after the reservoir variables. The whole system of equations is solved with Newton's method, i.e.:

$$\vec{\xi}^{(k+1)} = \vec{\xi}^{(k)} - \left[\left(\frac{\partial \vec{\psi}}{\partial \vec{\xi}} \right)^{(k)} \right]^{-1} \vec{\psi}^{(k)} \quad (\text{D12.2})$$

where

$$\left(\frac{\partial \vec{\psi}}{\partial \vec{\xi}} \right)$$

is the Jacobian matrix, and the superscript (k) denotes the iteration level. The Jacobian matrix in Equation (D12.2) is a sparse matrix that is solved by Incomplete LU (ILU) factorization followed by the GMRES iterative method. As the phase equilibrium Equation (D2.1), the chemical equilibrium Equation (D6.1) and the mineral material balance Equation (D10.3) do not involve variables in adjacent grid blocks, they are eliminated in a pre-processing step prior to applying the ILU factorization (Nghiem and Rozon, 1988 and 1989).

The above solution method allows the simultaneous convergence of the flow equations, the phase equilibrium equations, the chemical equilibrium equations and the mineral dissolution and precipitation rates. It is the most robust approach for solving these systems of strongly coupled nonlinear equations.

